

IBM Data Analyst Capstone

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- Database Trends – Findings & Implications
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EXECUTIVE SUMMARY



- Analyzed Survey Data
- Built Interactive Cognos Dashboard
- Identified trends in current tech, future tech interests, and respondent demographics
- Summarized findings to support data driven decision making.



INTRODUCTION



Purpose: Analyze developer survey responses

- Explore tech adoption and preferences
- Identify future technology trends
- Examine respondent demographics
- Present findings using an interactive dashboard



METHODOLOGY

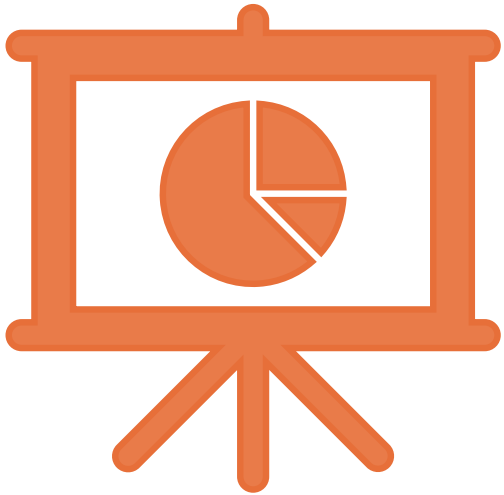


Data source: Imported survey dataset

- Cleaned missing and inconsistent values
- Performed EDA
- Created visualizations via python/Jupyter
- Built interactive dashboards in Cognos
- Summarized Key Insights



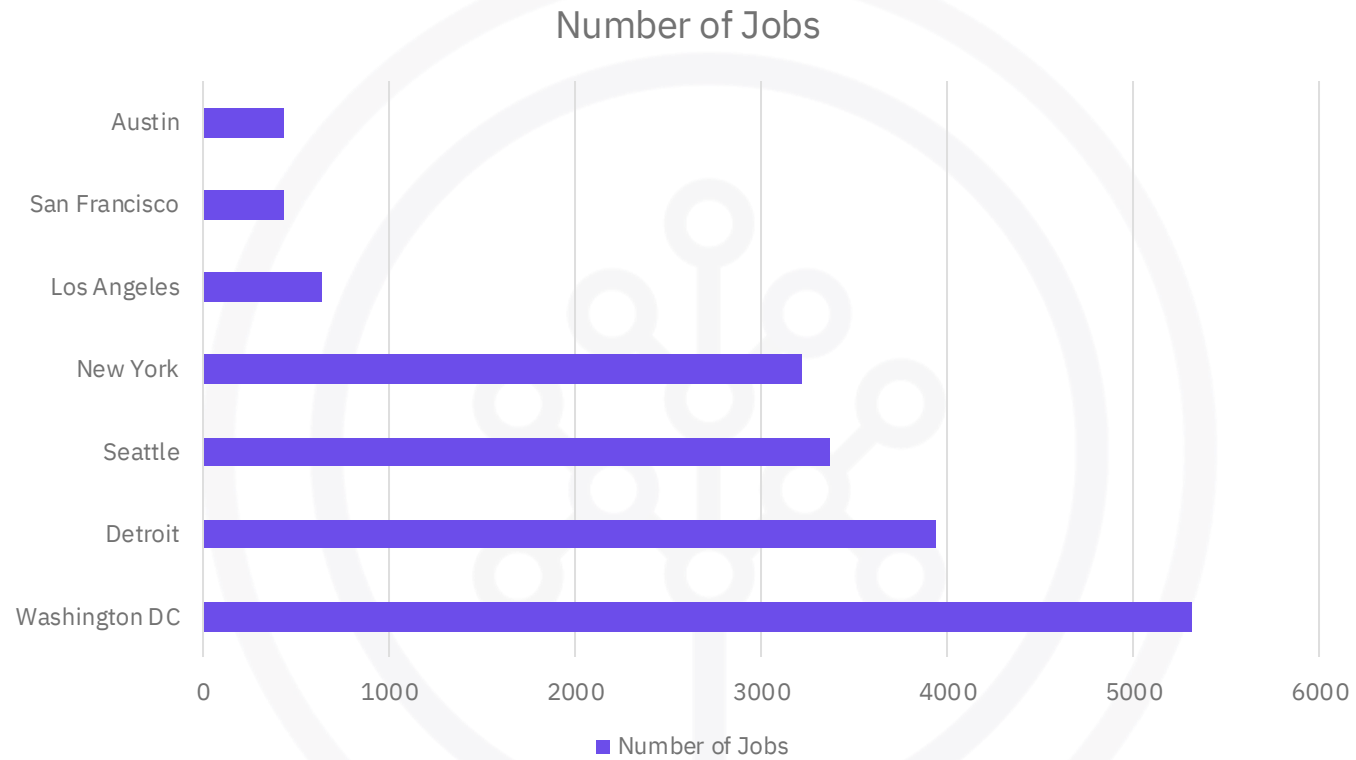
Results



The following slides summarize the Key findings from the given survey data using Python and IBM Cognos dashboards.



JOB POSTINGS

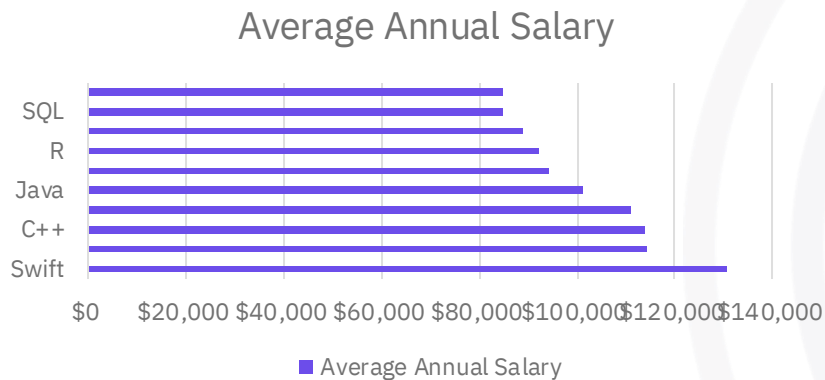


POPULAR LANGUAGES



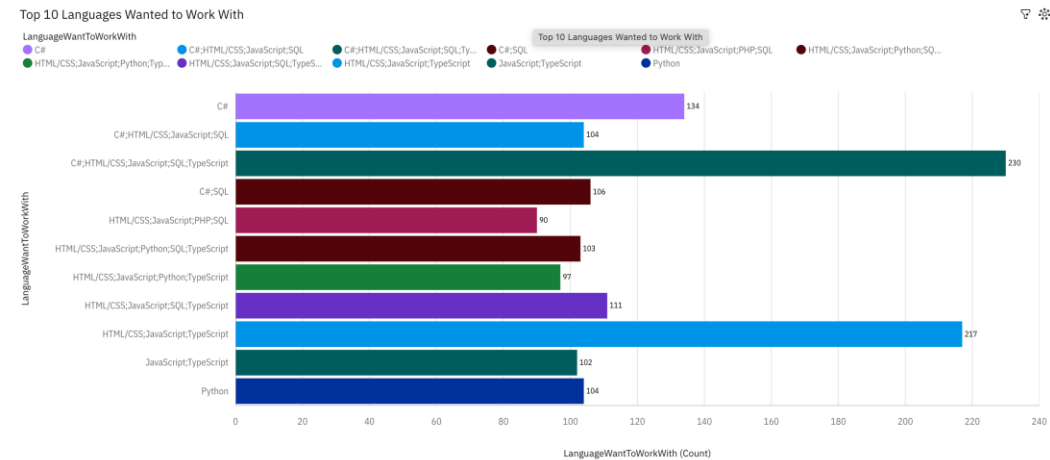
PROGRAMMING LANGUAGE TRENDS

Current Year



- Python, Swift, C++, and JavaScript are among the highest paying and most popular programming languages.
- Established languages such as Java, C#, SQL, and PHP remain widely used.

Next Year



- JavaScript, TypeScript, Python, SQL, and C# continue to rank among the most desired languages.
- Interest remains strong in modern web development and full stack technologies, suggesting continued demand for these skills.

PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- **JavaScript, Python, SQL, and C# remain among the most popular and in-demand programming languages.**
- **Interest in web development technologies such as TypeScript and modern frameworks continues to grow.**
- **Cloud platforms like AWS and Microsoft Azure remain highly desired skills among developers.**

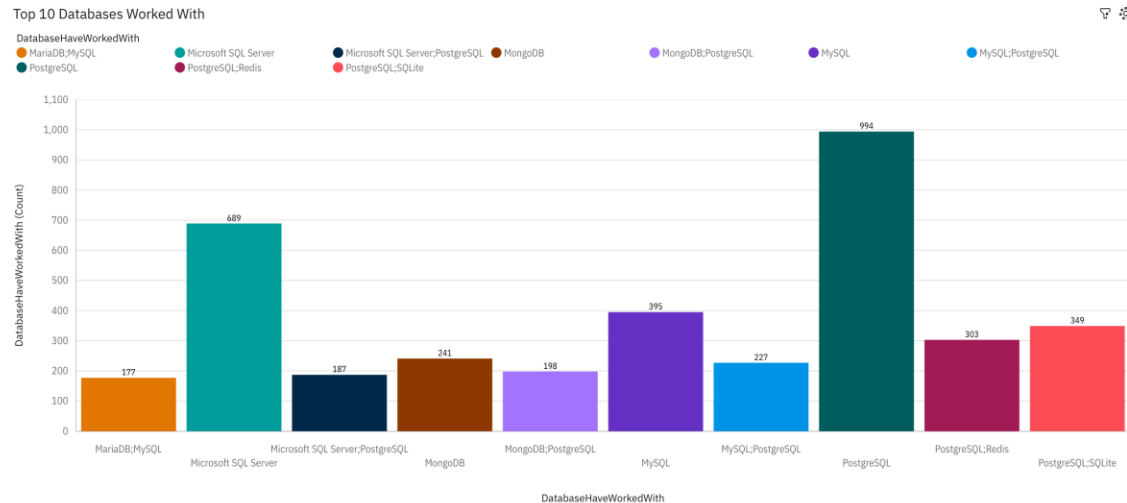
Implications

- **Developers should prioritize learning widely adopted languages and cloud technologies to remain competitive.**
- **Organizations should continue investing in modern web development tools and cloud infrastructure.**
- **Training programs should emphasize full-stack development, cloud computing, and database skills to meet future industry demand.**



DATABASE TRENDS

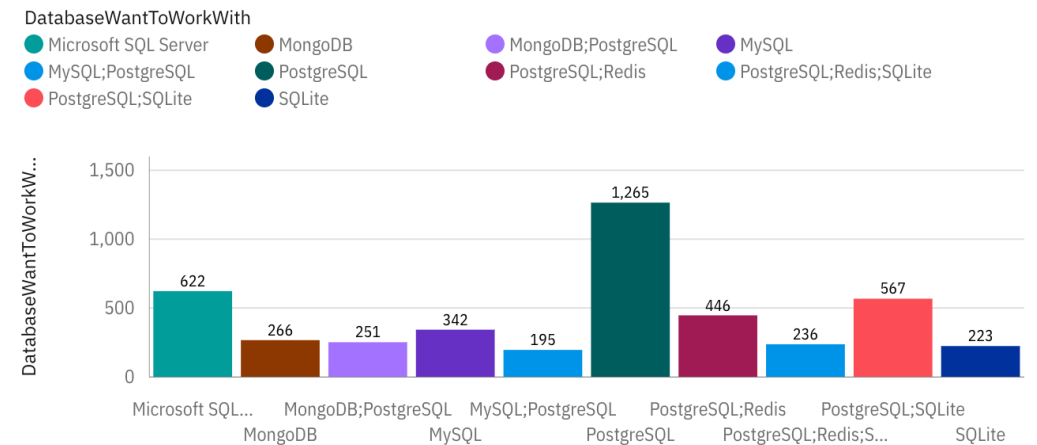
Current Year



- **PostgreSQL, MySQL, Microsoft SQL Server, and MongoDB are among the most widely used databases.**
- **Relational databases continue to dominate current development environments.**
- **Developers commonly use multiple database technologies depending on project requirements.**

Next Year

Top 10 Databases Wanted to Work With



- **PostgreSQL shows strong future demand and continued growth in popularity.**
- **Interest in PostgreSQL, MySQL, and MongoDB remains high for future projects.**
- **Developers are increasingly adopting modern open-source database technologies.**



DATABASE TRENDS - FINDINGS & IMPLICATIONS

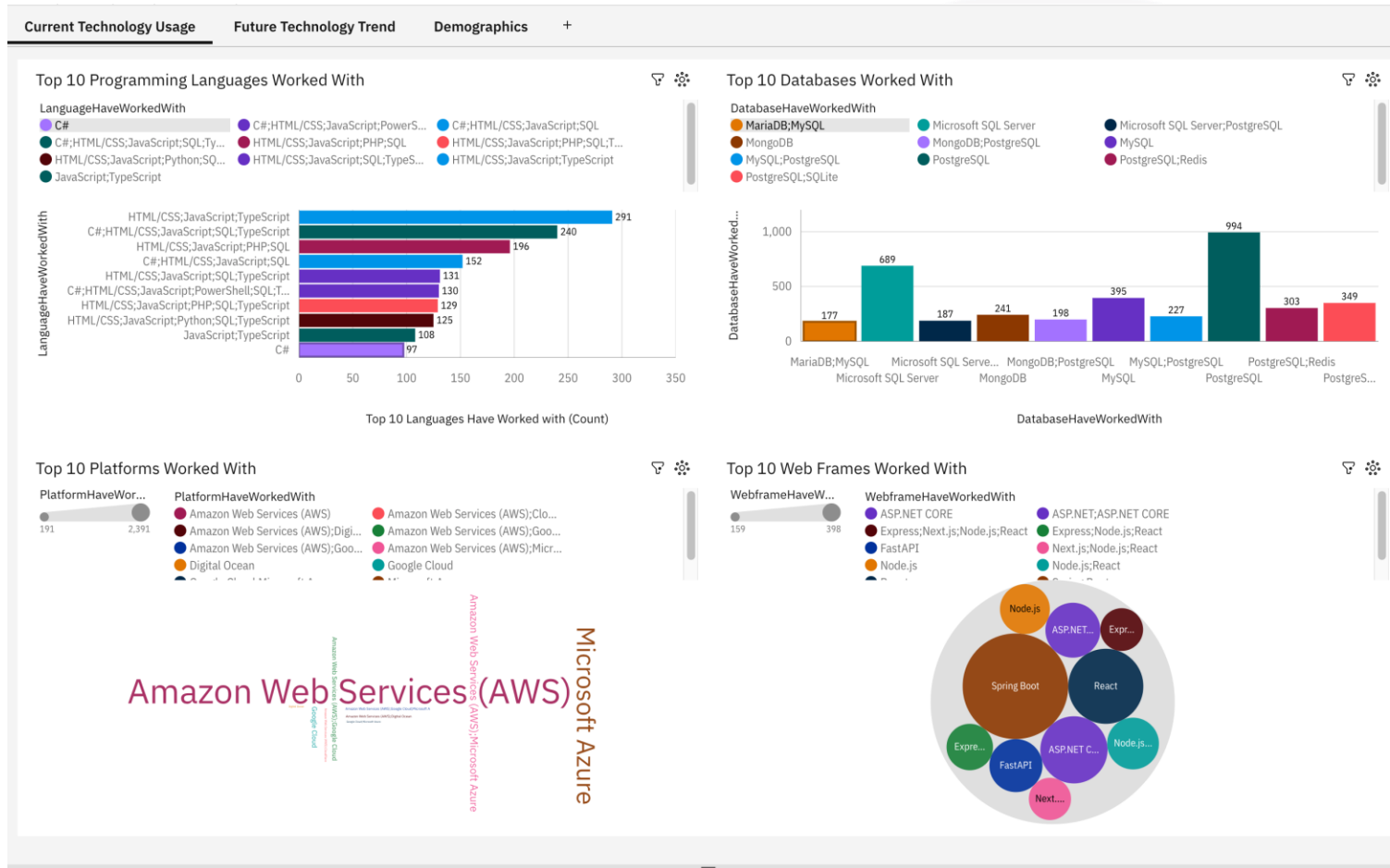
Findings:

- **PostgreSQL, MySQL, and Microsoft SQL Server remain the most widely used databases.**
- **Open-source databases continue to gain popularity among developers.**
- **Developers frequently work with multiple database technologies across projects.**

Implications:

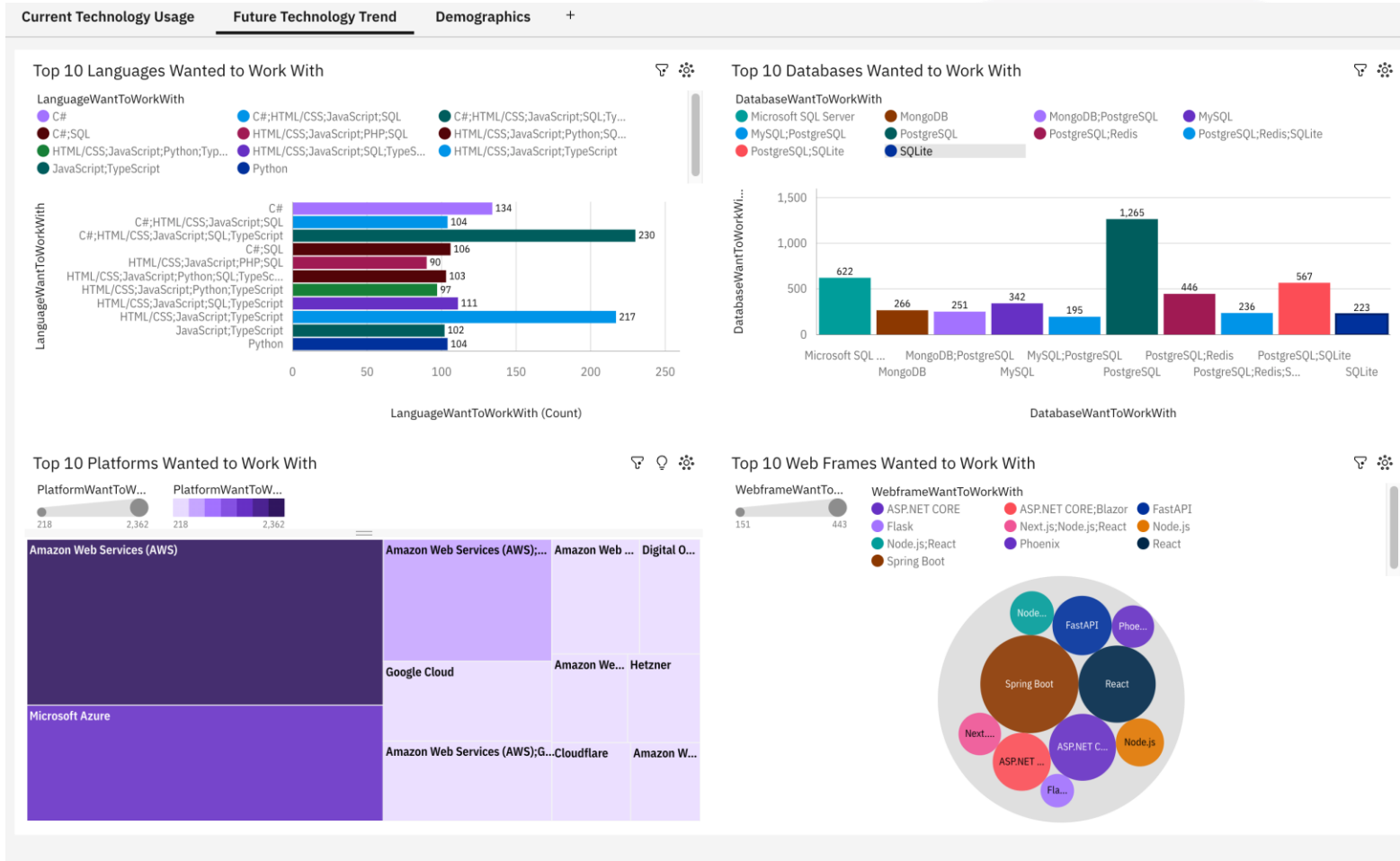
- **Organizations should continue investing in PostgreSQL and other open-source database solutions.**
- **Database professionals should develop skills in both relational and NoSQL technologies.**
- **Businesses should adopt flexible database strategies to support evolving application needs.**

DASHBOARD TAB 1: Current Technology Usage



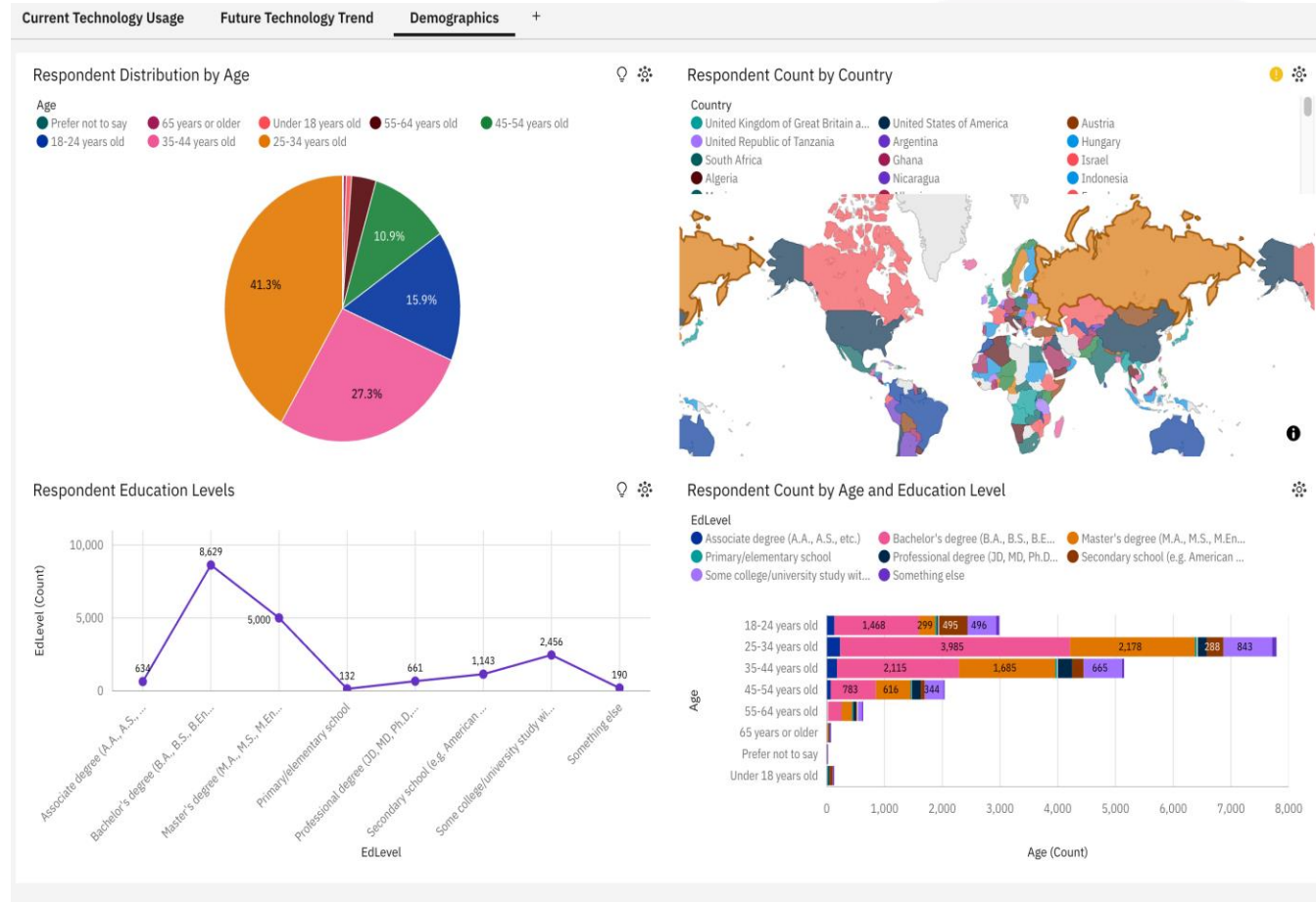
- **Modern web development dominates current technology usage.** Combinations centered around **HTML/CSS, JavaScript, TypeScript, SQL, and C#** appear most frequently among respondents.
- **PostgreSQL is the most commonly used database**, followed by **Microsoft SQL Server** and **MySQL**, highlighting the continued importance of relational database systems.
- **Cloud platforms are widely adopted**, with **Amazon Web Services (AWS)** and **Microsoft Azure** standing out as the most commonly used platforms.
- **React and Spring Boot are among the leading web frameworks**, indicating strong adoption of modern front-end and enterprise application development tools.

DASHBOARD TAB 2: Future Technology Trends



- **Developers continue to show strong interest in modern programming languages, with JavaScript, Python, TypeScript, SQL, and C#** remaining among the most desired technologies.
- **PostgreSQL is the most desired database technology**, suggesting it is expected to remain a leading choice for future development.
- **AWS and Microsoft Azure continue to dominate cloud platform interest**, indicating that cloud computing skills will remain highly valuable.
- **React and Spring Boot remain among the most sought-after web frameworks**, reflecting continued demand for modern front-end and enterprise application development.

DASHBOARD TAB 3: Demographics



- The largest group of respondents is aged **25–34**, representing roughly **41%** of survey participants, followed by the **35–44** age group at about **27%**.
- Most respondents hold a **Bachelor's degree**, while **Master's degrees** are the second most common level of education.
- The survey represents a **global developer community**, with respondents coming from countries across North America, Europe, Asia, Africa, and South America.
- The **25–34 age group** also has the **highest number of Bachelor's and Master's degree holders**, suggesting this demographic represents a large portion of the active professional developer workforce.



DISCUSSION



- **Full-stack technologies remain the most widely used**, with JavaScript, HTML/CSS, SQL, and TypeScript appearing across many of the most common technology combinations.
- **Interest is shifting toward modern cloud and web technologies**, particularly AWS, Azure, React, Spring Boot, and PostgreSQL.
- **The survey population is primarily composed of early- to mid-career developers (ages 25–44)**, with most respondents holding Bachelor's or Master's degrees.
- **Cloud platforms and modern databases continue to grow in importance**, indicating organizations are investing in scalable, cloud-native applications.
- **Overall, the results suggest that developers who build skills in JavaScript ecosystems, SQL/PostgreSQL, cloud platforms, and modern web frameworks will be well positioned for future opportunities.**



OVERALL FINDINGS & IMPLICATIONS

Findings:

- **Modern web technologies dominate** developer usage and future interest.
- **Cloud platforms and PostgreSQL show strong growth**, indicating continued adoption of cloud-native development.
- **Most respondents are early- to mid-career professionals** with Bachelor's or Master's degrees.

Implications:

- **Developers should prioritize JavaScript, SQL/PostgreSQL, cloud platforms, and modern frameworks** to remain competitive.
- **Organizations should continue investing in cloud infrastructure and modern application stacks** to align with industry trends.
- **Training and hiring efforts should focus on in-demand technologies** that are widely used today and expected to grow.



CONCLUSION



- **Modern web development technologies remain the most widely used and most desired skills.**
- **Cloud platforms (AWS and Azure) and PostgreSQL are among the fastest-growing technologies.**
- **The survey indicates a young, highly educated developer workforce driving technology adoption.**
- **Professionals who build skills in JavaScript, SQL, cloud computing, and modern frameworks will be well positioned for future career opportunities.**



APPENDIX



- No additional supporting charts or tables were required for this analysis.

